MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 100TH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode

See figure 1

Weight: 4-ounces (113.4 grams) nominal

Mounting position: Vertical

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ec	Ib	Ic	Pp
Unit:	MHz	V	Vdc	Vdc	mAdc	mAdc	W
Maximum:	40	5.25	3,000	-210	225	50	100
Minimum:	---	4.75	---	---	---	---	---
TEST CONDITIONS:	---	5.0 Vac	2,500	Adjust	40	---	---

GENERAL:

Qualification - Required

© denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:47Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	5.8	6.6	Aac
1261	Electrode voltage (1) (grid)	-		0.65	II	Ec(1)	-34.0	-46.0	Vdc
1261	Electrode voltage (2) (grid)	-	Eb = 14.0 kVdc, Ec/Ib = 1.0 mAdc	0.65	II	Ec(2)	---	-550	Vdc
1266	Total grid current	1		0.65	II	Ic	---	-10	µAdc
1266	Primary grid emission	-	Ic = 100 mAdc, t = 15 seconds; Ef = 5.5 Vac	0.65	II	Isg	---	-500	µAdc
1231	Peak emission	-	eb = ec = 2,500 v	0.65	II	is	3.0	---	a
	Quality conformance inspection, part 2								
1316	Amplification factor	-		---	---	Mu	34.0	42.0	---
1236	Power oscillation	-	F = 8 MHz; Eb = 3,000 Vdc; Ib = 150 mAdc	---	---	Po	270	---	W (useful)
1331	Direct-interelectrode capacitance	-		---	---	Cgp	1.7	2.3	pF
						Cin	2.5	3.4	pF
						Cout	---	0.45	pF
	Quality conformance inspection, part 3								
(C) 1031	Low-frequency vibration	2	No voltages applied	---	---	---	---	---	---
(C) 1036	Bump	2	Angle = 10°	---	---	---	---	---	---
---	Life test	-	Group C; power oscillation; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission	-		---	---	is	2.5	---	a
1266	Primary grid emission	-		---	---	Isg	---	-500	µAdc

NOTES:

1. This test shall be the first test performed at the conclusion of the holding period.

(C)2. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

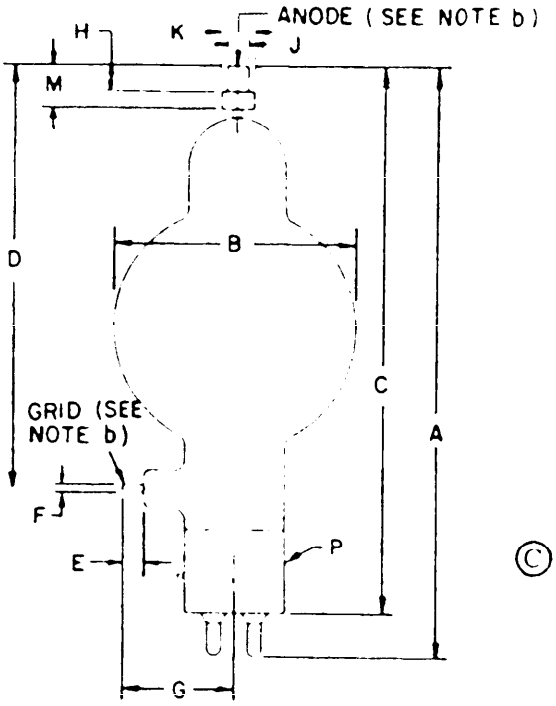
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

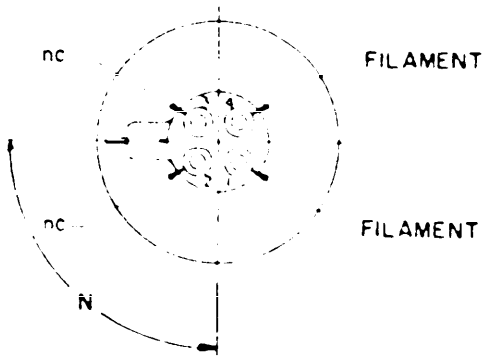
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2941)

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:47Z
Check the source to verify that this is the current version before use.





Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	7.250 (184.15)	7.750 (196.85)
B	---	3.188 (80.98)
D	5.000 (127.00)	5.500 (139.70)
E	.250 (6.35)	---
G	1.250 (31.75)	1.500 (38.10)
N	80	100
© Quality conformance inspection, part 3 (see note a)		
C	6.625 (168.28)	7.125 (180.98)
F	.055 (1.40)	.061 (1.55)
J	.350 (8.89)	.365 (9.27)
P	Base: A4-10 (EIA)	
Reference dimensions		
H	.328 (8.33)	
K	.500 (12.70)	
M	.563 (14.30)	

NOTES:

a. These dimensions shall be checked annually with the following sampling plan:

n1 = 4 c1 = 0 where c2 represents the total allowable failures
n2 = 4 c2 = 1: for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 basis.

© b. Heat radiating connectors HR-2 (grid) and HR-6 (anode) or equivalent, are recommended for all applications.

FIGURE 1. Outline drawing of electron tube type 100TH.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:47Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PA
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:47Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:47Z
Check the source to verify that this is the current version before use.